

$^{32}\text{S}(^3\text{He}, n\gamma)$ 1974Gr19, 1974Be18, 1972Ca22

1974Be18: an 8-MeV ^3He beam was provided from the ANU EN tandem accelerator and impinged on cooled sulphur targets.

Neutrons were detected using an NE213 liquid scintillator and γ rays were detected using a Ge(Li) detector and a NaI(Tl) detector. Measured E_γ , I_γ , $n\gamma$ -coin, $n\gamma(\theta)$. Deduced levels, J, γ -ray branching ratios; deduced lifetimes for 4 levels using DSAM.

1974Gr19: 8-12-MeV ^3He beams were provided from the 7-MV Van de Graaff accelerator at the University of Freiburg and impinged on natural ZnS targets. Neutrons were detected using an annular NE213 liquid scintillator and γ rays were detected using a Ge(Li) detector and 5 NaI(Tl) detectors. Measured E_γ , I_γ , $n\gamma$ -coin, and $n\gamma\gamma$ -coin. Deduced 10 levels, J, γ -ray branching ratios, and mixing ratios; deduced lifetimes for 5 levels using DSAM.

1972Ca22: an 8.5-MeV ^3He beam was provided from the Oxford University tandem Van de Graaff generator and impinged on Ag_2S targets. Neutrons were detected using a NE102 liquid scintillator. The γ rays were detected using a Ge(Li) detector. Measured E_γ , I_γ , E_n , I_n , and $n\gamma(\theta)$. Deduced lifetimes for 3 levels using DSAM.

 ^{34}Ar Levels

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2}</u>	<u>Comments</u>
0.0	0 ⁺		
2091.0 4	2 ⁺	139 fs 35	J ^π : from the Adopted Levels. Spin=2 confirmed by $n\gamma(\theta)$ in 1974Gr19. T _{1/2} : from lifetime $\tau=200$ fs 50: weighted average of 330 fs 80 (1974Be18, DSAM), 200 fs 60 (1974Gr19, DSAM), and 150 fs 50 (1972Ca22, DSAM).
3287.5 6	2 ⁺	104 fs 42	J ^π : from the Adopted Levels. Spin=2 confirmed by $n\gamma(\theta)$ in 1974Gr19. T _{1/2} : from lifetime $\tau=150$ fs 60: weighted average of 215 fs 70 (1974Be18, DSAM), 130 fs 60 (1974Gr19, DSAM), and 120 fs 65 (1972Ca22, DSAM).
3873 3	0 ⁺	>416 fs	J ^π : from the Adopted Levels. Spin=0 confirmed by $n\gamma(\theta)$ in 1974Gr19. T _{1/2} : from lifetime $\tau>600$ fs (1974Gr19, DSAM).
4127.4 11	(2 ⁺)	<243 fs	J ^π : spin=2 from $n\gamma(\theta)$, but did not exclude spin=3 or 4; $\pi=+$ from mirror level (1974Gr19). T _{1/2} : from lifetime $\tau<350$ fs (1974Be18, DSAM). Other: $\tau>400$ fs (1974Gr19, DSAM).
4513.5 8	3	201 fs 55	J ^π : spin=3 from $n\gamma(\theta)$ (1974Gr19). T _{1/2} : from lifetime $\tau=290$ fs 80: weighted average of 390 fs 180 (1974Be18, DSAM), 260 fs 80 (1974Gr19, DSAM), and 300 fs 80 (1972Ca22, DSAM).
4631 [‡] 4			
4865 [‡] 4			
4967 [‡] 4			
5255 [‡] 4			
5542 [‡] 4			

[†] For E(level) below 4.6 MeV, from a least-squares fit to γ -ray energies.

[‡] From 1974Gr19 based on γ data.

 $\gamma(^{34}\text{Ar})$

<u>E_i(level)</u>	<u>J_i^π</u>	<u>E_γ[†]</u>	<u>I_γ[‡]</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.</u>	<u>δ</u>	<u>Comments</u>
2091.0		2 ⁺	2090.9 4		100			0.0 0 ⁺
3287.5		2 ⁺	1196.5 4		89 5		2091.0 2 ⁺	M1+E2 +0.12 5 E _γ : weighted average of 1196.6 4 (1974Be18) and 1195.8 14 (1972Ca22). I _γ : weighted average of 92 5 (1974Be18), 90 7 (1972Ca22), and 85 5 (1974Gr19). Mult., δ: D+Q from $n\gamma(\theta)$ (1974Gr19); M2 ruled out by RUL.
			3286 4		11 5		0.0 0 ⁺	A ₂ =+0.56 10 (1974Be18). E _γ : from 1974Be18. I _γ : weighted average of 8 5 (1974Be18), 10 7 (1972Ca22), and 15 5 (1974Gr19).

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$^{32}\text{S}(^3\text{He}, n\gamma)$ **1974Gr19, 1974Be18, 1972Ca22 (continued)** $\gamma(^{34}\text{Ar})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
3873	0^+	585		<30		3287.5 2^+ I_γ : <30 (1974Be18).
		1782.3		>70		2091.0 2^+ E_γ, I_γ : from 1974Be18; other: $I_\gamma=100$ (1974Gr19).
4127.4	(2^+)	839.9	9	90	5	3287.5 2^+ E_γ : weighted average of 840.1 9 (1974Be18) and 839.1 18 (1972Ca22). I_γ : others: >50 (1974Be18), 100 (1972Ca22). δ : +0.52 15 from $n\gamma(\theta)$ if $J_i=2$. I_γ : other: <50 (1974Be18).
		2037		10	5	2091.0 2^+
		4128		<10		0.0 0^+
4513.5	3	1226.0	6	90	10	3287.5 2^+ E_γ : weighted average of 1225.5 6 (1974Be18) and 1226.7 7 (1972Ca22). I_γ : others: >90 (1974Be18), 100 (1972Ca22). $A_2=-0.38$ 24 (1974Be18). I_γ : others: <10 (1974Be18), <20 (1972Ca22).
		2422		10	3	2091.0 2^+
4631		2540		>50		2091.0 2^+
4865		2774		>50		2091.0 2^+
4967		841		<20		4127.4 (2^+)
		2876		>50		2091.0 2^+
5255		3164		>50		2091.0 2^+
5542		911		35	10	4631
		1029		65	10	4513.5 3

 † E_γ values without uncertainties are from level-energy differences. ‡ From 1974Gr19, unless otherwise noted.

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Level Scheme

Intensities: % photon branching from each level

